ELECTRICAL—FALSE ALARM ACTIVATION—ALARM SYSTEM INOPERATIVE—SERVICE TIPS	Article No. 02-4-6

MERCURY: 1999-2000 COUGAR

ISSUE

Some vehicles may exhibit complaints of false alarm activation of the Perimeter Anti-Theft Alarm or of the alarm being inoperative. This may be due to any of the following:

- Incomplete understanding of how to operate the alarm system
- System operating characteristics
- Component fault

ACTION

This reference material is intended to supplement existing Workshop Manual procedures by providing Service Tips to improve diagnostic capabilities.

SERVICE INFORMATION

Anti-Theft System Overview

The Perimeter Anti-Theft System is present on all Cougar's equipped with the Remote Keyless Entry (RKE) system. It is not possible to equip the Cougar with only the Perimeter Anti-Theft System or only RKE separately. This is an "active" system: it does not arm automatically, but instead must be actively armed

All Cougars are equipped with an engine immobilizer called the Passive Anti-Theft System (PATS), and whenever the Perimeter Anti-Theft System senses an unauthorized entry, it will cause the PATS lamp (on the instrument panel) to stay on constantly. However the anti-theft system does not directly affect the PATS or Electronic Engine Control (EEC) system and will not result in a no crank or no start condition!

The Major Components Of The System Are:

Central Security Module (CSM): This is located in the lower RH side of the instrument panel, behind the glove box.

Trigger Inputs/Switches: A change in the state of any of these inputs/switches from open-to-closed or from closed-to-open will be interpreted by the CSM as an unauthorized activity and will result in the alarm system being triggered. The following components are monitored once the system is actively set/armed:

- LH Door
- RH Door
- Lift gate
- Hood
- Radio

Alarm System Horn: This is located in the RR quarter panel. The under hood horn is not utilized by this system.

Lift gate Inhibit Switch: This switch is operated by the lift gate key cylinder. The switch temporarily disables the system when the key is used to unlock the lift gate, thereby preventing activation of the alarm. Lift gate and hood ajar switches cannot trigger an alarm when the lift gate is opened with the key. The other alarm inputs (LH Door, RH Door and Radio) will still cause an alarm when activated. The system will be armed again 20 seconds after the lift gate is closed.

NOTE

IF THE LIFT GATE KEY CYLINDER IS TURNED TO OPEN, THE LIFT GATE AJAR SWITCH WILL NOT TRIGGER AN ALARM WHEN THE LIFT GATE IS OPENED WITHIN 20 SECONDS AFTER TURNING THE LIFT GATE KEY CYLINDER. IF THE LIFT GATE IS HELD DOWN FOR MORE THAN 20 SECONDS AFTER HAVING TURNED THE LIFT GATE KEY CYLINDER SWITCH AND THEN THE LIFT GATE IS OPENED AFTER THESE 20 SECONDS, THE ALARM WILL BE TRIGGERED.

Once the Perimeter Anti-Theft System activates (signal lamps flashing and/or horn sounding) it can only be deactivated by:

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1. Pushing the "UNLOCK" button on the RKE Transmitter. Pushing "PANIC" WILL NOT deactivate the system.It will only start a Panic Alarm on top of the Perimeter Alarm
2. Unlocking either front door with the key.
3. Turning the ignition switch to the "on" position (vehicles produced after August 2000).

A "Panic" feature is also provided with the RKE system. Although this shares many system components and apparent operational functions, it is essentially a completely separate system from the anti-theft system and should be diagnosed accordingly.

When "PANIC" is pushed on the RKE transmitter, the signal lamps will flash and the interior alarm will sound. This operational mode can only be canceled by:

- Pressing the "PANIC" button on the RKE a second time
- Inserting the key into the ignition and turning it to the "on" position

ARMING

After setting the system, there is a 20 second delay until the system is armed. The PATS LED flashes when armed. Any change to the trigger inputs/switches will not result in the alarm system being activated until after this 20 second period has lapsed. The 20 second delay until the alarm system is armed will be restarted after every trigger input/switch (except radio) is back in "closed" (door is closed) state.

Arming With The RKE Key FOB (Remote Transmitter)

1. The key must be out of the ignition.
2. RH and LH doors, Lift gate, and Hood must be closed.
3. One push of the "lock" button will set/arm the system. A visual flasher indication will confirm that LH Door, RH Door, Lift gate and Hood are closed and the alarm system will be armed after 20 seconds.
4. Pushing the "lock" button a second time will result in a visual confirmation that the system is armed:
   a. The front/rear turn signal and side marker lamps will flash 1 time.
   b. On vehicles built after February 2000 the alarm system horn will also chirp one time (to confirm that the vehicle is centrally locked).

NOTE
IF THE SIGNAL LAMPS DO NOT FLASH BUT THE HORN CHIRP OCCURS, THEN ONE INPUT (RH DOOR, LH DOOR, HOOD, OR LIFT GATE) IS OPEN. THE SYSTEM WILL ARM 20 SECONDS AFTER CLOSING.

NOTE
THIS ALSO APPLIES TO ARMING WITH THE DOOR KEY. WHEN ARMING WITH THE DOOR KEY, THERE WILL NOT BE AN OPTICAL FLASHER OR ACOUSTICAL HORN CHIRP FEEDBACK.

Arming With The Interior Power Lock Switch

1. The key must be out of the ignition.
2. The door must be open.
3. Push the "lock" button on the interior trim panel and close the door afterwards.
4. There will be no visual (lights) or audio (horn) confirmation that the system is now set/armed.

Arming With The Interior Manual Lock Lever

The manual lock lever beneath the interior door latch lever cannot be used to arm/set the alarm system.

Arming With The Door Lock Key

1. The key must be out of the ignition.
2. Both doors must be closed.
3. Lock the drivers or passenger door.
4. There will be no visual (lights) or audio (horn) confirmation that the system is now set/armed.

OPERATING STRATEGIES

Once triggered by an apparent unauthorized entry, after 30 seconds of operation the horn will stop. However the exterior signal lamps will continue to flash for 4.5 minutes (total flashing time is 5 minutes).

1. While the exterior lamps are flashing (but the horn is off) or the exterior lamps have stopped flashing, creating another unauthorized entry (e.g. reopening a door) will result in the horn being activated for another 30 seconds. Also the exterior signal lamps will start flashing for another 5 minutes.

    a. During the time period that the horn is active, unauthorized entry will not result in a new alarm. Only after the 30 seconds horn activation time has elapsed, a new alarm can be triggered.

    b. Pressing "UNLOCK" on the RKE transmitter or unlocking the driver or passenger door with a key will deactivate the alarm. The system will not be set/armed until actively set/armed by the customer.

2. On vehicles built prior to August 2000, if after entering an unlocked vehicle, the driver locks the doors (with remote control or with the Interior Power Lock Switch) and waits over 20 seconds before starting the engine, the alarm will sound when turning the ignition switch to the "on" position (by design). The system can then be disarmed using either the "UNLOCK" button on the RKE, or by unlocking a door with the key. This is a normal operating strategy of the system and is designed to enhance system security.

3. When the system is armed the interior door lock switches are disabled, by design. However, the manual door locks can still be operated.

DIAGNOSTIC MODE

The CSM software contains a diagnostic mode, which can be used to verify that the proper signals are being received by the CSM. This can be especially useful in determining why the system will not set/arm, but under certain circumstances could also possibly result in customer complaints while driving.

To enter the diagnostic mode:

1. Turn the ignition key off.

2. The alarm system has to be disarmed.

3. Actuate one of the vehicle's ajar switches (RH door, LH door, hood or lift gate) 7 times in a row within 8 seconds (by opening and closing the door).

    a. The horn and exterior signals will be activated briefly to confirm the diagnostic mode has been initiated after actuating the ajar a 7th time.

NOTE
THE HOOD SWITCH IS EASILY ACCESSIBLE IF YOU ARE ACTUATING THE SWITCH BY HAND.

While in the diagnostic mode, anytime the following trigger inputs change state (from inactive to active) the horn and exterior signal lamps will be activated:

1. Open Hood.

2. Open Lift gate, or Turn Anti theft inhibit switch.

3. Open Driver or Passenger Door.

4. Interior power lock switch (pressing lock or unlock position).

5. Lock/Unlock switch in the driver and passenger door key cylinder.

6. Taking out the radio.

7. Turning the ignition to the "on" position.

8. Pressing the internal lift gate release button.

9. Alarm set/reset switches.

10. Pressing any RKE button (also an unprogrammed FOB)

The system will automatically exit the diagnostic mode if it doesn't see a change in state of any one of the trigger inputs within 20 seconds. Exiting the diagnostic mode will be confirmed by a short activation of the horn and the exterior signal lamps.

NOTE
IF THE SYSTEM IS IN THE DIAGNOSTIC MODE AND THEN THE IGNITION KEY IS TURNED ON, IT IS POSSIBLE THAT IT STAYS IN THIS MODE EVEN WHILE DRIVING. AS A RESULT, IF A TRIGGER INPUT (SUCH AS A DOOR/LIFT GATE AJAR SWITCH DUE TO IMPROPER ADJUSTMENT) IS CYCLED WHILE DRIVING, THE EXTERIOR SIGNAL LAMPS AND HORN WILL BE ACTIVATED.

Continuous monitoring of PIDS can be done through the Hybrid Electronic Cluster (HEC) module. Refer to section 413-01 in the Workshop Manual.

CUSTOMER INQUIRY

When diagnosing any suspended Perimeter Anti-Theft System or "Panic" system concern, obtaining a clear description of the concern is vital. Whenever possible, try to determine:

What is specifically happening?

1. The exterior signal lamps (only) flash.
2. The horn (only) is activated.
3. The exterior lamps and horn are activated, but the horn eventually stops while the lamps continue to flash.
4. Which button on the RKE transmitter will deactivate the system inadvertently activated?
5. There is no flasher feedback when arming the system with the lock button on the remote control.

When does the concern occur?

1. When locking or unlocking the doors.
2. When opening the door(s).
3. When turning on the ignition.
4. While driving.
5. While parked and unattended.
6. While parked and attended (leaning on the finder, washing the car, etc.)

What is the PATS Lamp Doing?

1. Off
2. Flashing
3. On Constantly

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 201000, 201200, 203000, 203200, 205000, 206000